# CS 830 Project Project report & results

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**Goal:** 2
**Why to evaluate inconclusiveness for Explicit reasoning tasks?** 2
**Problem statement** 5
Word Sense Disambiguation: 6
Prompt design follows Chain-of-thought prompt design used in [7]. 6
There are two KBs: 7
1) KB 1 7
2) KB 2 7
**Input prompts with knowledge base:** 7
**Method summary:** 10
Detailed Method Summary: 10
**Output (same as output format from assignment 7):** 10
**ClaudeAI and ChatGPT** 11
**Intuition and Motivation behind this problem (based on review comments):** 11
**First Order Logic as a language to represent Natural Language:** 11
**Results:** 11
Preliminary results for designing relevant, consistent and specific CoT prompts 1,2,3 and 4 : 12
Final results for prompts 5,6,7 and 8: 12
**Observations:** 13
**Key Observations and questions/potential answers:** 14
**Time and Space complexity, State Space** 16
**How to generate a really long knowledge base and convert to Skolemized clausal forms?** 17
**Example Knowledge Bases, Clausal Forms and proofs** 18
Example Knowledge base 1: 18
Skolemized Clausal forms (manual): 18
Knowledge base 2 (with noise): 20
Skolemized Clausal forms (manual): 21
**References** 22
**Appendix** 24
Example problem to explain context for why conventional skolemized normal forms are NOT Meaning-preserving: 24
Selecting example of Tax-cut problem from [4] : 24

First Order Logic form: 24
CNF form: 25
Conventional Skolemization: 25
Resolution Refutation Solution using Conventional Skolemized Causal forms: 25

## Goal:

Can Large Language models (LLMs) say "inconclusive" given a knowledge base by using resolution refutation, despite its word sense disambiguation accuracy and world knowledge? LLMs hallucinate and are sensitive to prompts [7] but can they really deduce a query as inconclusive, given a Knowledge base?

## Why to evaluate inconclusiveness for Explicit reasoning tasks?

Do LLMs respond with "I don't know" or "inconclusive"? LLMs detect ambiguities during mathematical reasoning. We often answer "I don't know" to express lack of knowledge. "This is Inconclusiveness" is a response that demonstrates that conclusion cannot be arrived at. Although the two responses are different, they are crucial and important parts of reasoning. I was trying to find use cases for inconclusive responses. The importance of lack of conclusion seems crucial to reasoning, in understanding what is missing or is ambiguous, uncertain and indeterminate. The lack of conclusion can help

Resolution is refutation complete, if a set of clauses, that are in first order skolemized clausal normal forms, are unsatisfiable, then the resolution derives empty clauses and verifies unsatisfiability. But Resolution is NOT decidable if the set of clauses may be satisfiable, i.e. if there is a query that is a logical consequence of the set of clauses, then the Resolution Refutation procedure may never terminate. If C is a first order clausal normal form that is unsatisfiable, then $C \vdash \perp$, since resolution combined with skolemization and classification is a refutation complete calculus for First Order Logic.**[11]** By Lifting lemma and Herbrand's theorem, proof for First-order resolution's refutation completeness can be proved. If a formula is satisfiable, proof search can terminate or go one forever, as there are infinitely many clauses that can be generated by resolution. Therefore resolution is not a decision procedure for First order logic and is undecidable if the program never terminates if the query is satisfied. There may be other forms of resolution **[12]** but we only use conventional skolemization decomposition for this project. The conventional skolemization representation/decomposition(CSD) are not meaning-preserving clausal forms. **[13]** Please refer to the Appendix section for an example where meanings are not preserved when using First order logical forms, involving existential quantifiers.
Resolution Refutation do not have world knowledge and do not consider word sense disambiguation.

Large Language Models (LLMs) have shown remarkable reasoning and semantic understanding by effectively performing on tasks such as Natural Language Understanding [23]. Large Language models have been trained on various types of languages such as Natural languages, and formal languages such as Java, C, and Python [19]. More recent GPT-4o [18] can effectively process the textual, visual, and audio inputs. The Large Language Models that are trained on large amounts of data that learn distributional semantics [25] perform moderately well on the basic commonsense reasoning tasks requiring analogy and logic, still require explicit reasoning and neurosymbolic methods, graph-based methods, and Retrieval Augmented Generation (RAG) based methods to improve the performance on analogy and logic [23].
The rules of logic are dependent on statistical patterns to conduct reasoning and Language models learn statistical features that inherently exist in the reasoning examples. **[25][26]** Vision-based Large language models on the other hand tend do better in Multimodal AI tasks such as Visual Question Answering (VQA), Vision Language Retrieval (VLR), and Natural Language for Visual Reasoning (NLVR) than the other Multimodal AI systems [21]. Aside from the computational costs and the api costs of applying LLMs to domain-specific tasks, the explicit-reasoning tasks still require hand-crafted domain-specific rules and logic that are difficult to scale and generalize to other applications [23]. The work [24] explores reasoning about how LLMs obtain answers and how they have difficulty during deductive reasoning. LLMs on the other hand, terminate and don't get stuck whether to conclude true, false or inconclusive, despite various other issues surrounding accuracy and planning during deductive reasoning. [7][15]

Large Language models (LLMs) serve as knowledge bases with world knowledge. [9] Large Language models do well in word sense disambiguation. [8] LLMs such as ChatGPT, GPTx models are able to translate natural language to first order logic and skolemized clausal normal forms.[4][7] LLMs are able to do deductive reasoning but don't do well in planning [7][15]. LLMs are known to have emergent abilities but recent works show evidence that emergent abilities are not fully emergent but are a combination of in-context learning, model memory and linguistic knowledge [16]. More recent works on canonical versus non-canonical tokenizations demonstrate exponential growth of number of tokenizations as a function of sentence length[17]. The work on tokenization in LLMs [17] explores the fact that the LLMs are shipped with tokenizers that may deterministically encode text in canonical tokenization that are assigned with probabilities. The tokenization is not unique in LLMs i.e. the same word [Bat] can be tokenized as [B,at], [B,a,t], [Ba,t], [Bat]. The probability of [Bat] is the probability of its canonical sequence of tokens. Such a behavior is inherent to autoregressive LLM. [17]. The merging of tokens happens as follows: first word is split into characters and priority rules

are applied. For example if each pair of characters in [Bat] generated, then a pair with maximum frequency of occurring together is chosen as a merge rule. This process seems to be repeated until desired vocabulary size is achieved, which is the Byte pair encoding (BPE). [17]. Merge rules are also dropped during training, while the rest of BPE is preserved. [17]. For each string x, there is a canonical tokenizer v* then canonical probability is p(x,v*). Determining by computing most likely canonical tokenization is hard as distributions of tokenizations of the word tokens shows that canonical tokens have higher marginal probabilities [17]. This seems to emphasize the exponential increase in search space with that of canonical tokenization. The inference time of an LLM is when an input text is provided to an LLM for which the time LLM takes to tokenize and generate vector embeddings and process to generate an output. Thus [17] argues choice of canonical versus non-canonical can significantly guide the responses from an LLM and we can tap into conditioning such tokenization to guide LLM responses based on most-likely tokenization or marginal string probability. Two approaches that are explored in [17] show adjusting the marginal estimate over tokenizations that are not canonical and by fine tuning hyperparameter $\alpha$ that adjusts weights for canonical versus non-canonical estimates. This work shows canonical tokenization ([B,at], [B,a,t]...) is the most likely tokenization which is also very close to the marginal probability of the text summed over tokenizations itself ([Bat]).

Now to finally answer the answer question "why First Order Logic" - Aside from the computational costs and the api costs of applying LLMs to domain-specific tasks, the explicit-reasoning tasks still require hand-crafted domain-specific rules and logic that are difficult to scale and generalize to other applications [23]. The work [24] explores reasoning about how LLMs obtain answers and how they have difficulty during deductive reasoning. LLMs on the other hand, terminate and don't get stuck whether to conclude true, false or inconclusive, despite various other issues surrounding accuracy and planning during deductive reasoning. [7][15] The works from [24] FOLIO, Natural Language Reasoning with First-Order Logic stands as a challenge dataset for testing Natural Language to First Order Logic (FOL) translation for evaluating Large Language Models for reasoning with FOL.

To answer why Word Sense Disambiguation, I chose word sense disambiguation because there are words that can confuse the context when using Resolution refutation. But on the other hand, skolemized clausal normal forms are not meaning-preserving. To be more specific, I chose words that have word sense disambiguation as well as that the two words have the same part of speech, same spelling but have different meanings. To design knowledge, I chose two such words that have different meanings but have the same parts of speech. Resolution Refutation (RR) over skolemized clausal normal forms

would likely not preserve the meaning unless we explicitly provide more information about context, meaning and other details explicitly. In the absence of world knowledge and word sense disambiguation as a capability, RR would not be fully able to provide correct proof if a given query is true or not. LLMs on the other hand, have shown lack of explicit reasoning for tasks that require such explicit reasoning, although they serve as knowledge bases with world knowledge as well as with word sense disambiguation capabilities. I individually prove that each technique by itself cannot fully support the if the query is true or false or inconclusive. By bringing the best of both techniques - implicit reasoning in LLMs based on word sense disambiguation, in-context learning and the explicit reasoning using RR.

## Problem statement

For this project, I attempt to test reasoning and results with a knowledge base for using Word Sense Disambiguation i.e. detecting correct meaning of the word, when it has multiple possible meanings (polysemy). The problem I attempt to test is "can LLM say I don't know" or "inconclusive" given knowledge base by using resolution refutation, given its word sense disambiguation accuracy and world knowledge? I am using the resolution refutation method here since it has been shown that LLMs do well the translation from Natural Language to First Order Logic. [27] As discussed in the aforementioned section, LLMs show emergent-like behavior combined with linguistic and in-context memory, I would like to verify if the claims of translation from NL to FOL are reasonable. Given a knowledge base in natural language, generate first order logic forms, then convert to skolemized clausal normal forms using LLMs and evaluate the clausal forms by Resolution Refutation (naming it RR method) algorithm and by also testing the generated clausal forms as inputs to LLMs itself and verify the results and compare responses/results between LLM and RR. Amidst the hallucinations and other problems LLMs run into, LLMs do seem to show signs of containing intrinsic representations. [28]

The problem of exploring inconclusive is interesting since LLMs may or may not necessarily restrict to given KB and interpretations within the scope of the KB. On the other hand, it may be required to test if they are able to detect to say "I don't know" or "inconclusive".

Research questions:

- Do LLMs really use resolution refutation for the clausal forms they generated from Natural Language Knowledge bases?
- DO LLMs run Resolution refutation to derive the proof for resolution?

- Are LLMs able to "decide" a satisfiability of query using Resolution Refutation alone?
- Do LLMs "decide" a satisfiability of a query with word sense disambiguation and world knowledge outside of the given KB using Resolution Refutation alone?
- Do LLMs "decide" a satisfiability of a query with word sense disambiguation and world knowledge outside of the given KB WITHOUT using Resolution Refutation alone?
- Do LLMs do better with best of both: First Order Logic combined with natural logic and common sense reasoning together?

I created two example knowledge bases in natural language that can have multiple First Order Logic (FOL) interpretations. I generated clausal forms from Large Language Models in this case I used Claude i.e. claude-3.5-haiku20241022 & claude-2.7-sonnet-20250219 versions respectively and chatGPT. (An FOL interpretation is an assignment of meaning to the symbols of a formal language.). This is unlike assignment 6 and 7, since clausal normal forms (CNFs) are generated by Large Language models (LLMs) given a natural language knowledge base and those clausal forms serve as inputs to resolution refutation as well as they serve as input prompts back to the LLMs.

## Word Sense Disambiguation:

**Example1:** Bat is a mammal. Bat is used in baseball.
**Example2:** Bark is a boat. Bark is a sheet of chocolate.
Domain of discourse: All words that have two different meanings {bat1 , bat2, bark1, bark2} where the meaning of each word is different. bat1 != bat2, bat1 != bark1, bat1!=bark2 and so on.

## Prompt design follows Chain-of-thought prompt design used in [7].

Q: Each cat is a carnivore. Every carnivore is not herbivorous. Carnivores are mammals. All mammals are warm-blooded. Mammals are vertebrates. Every vertebrate is an animal. Animals are multicellular. Fae is a cat. True or false: Fae is not herbivorous. A: Fae is a cat. Cats are carnivores. Fae is a carnivore. Every carnivore is not herbivorous. Fae is not herbivorous. True (Saparov A, He H 2023.)

## There are two KBs:

### 1) KB 1

Knowledge base is as follows:
Bat is a mammal. Bat is used in baseball. Bats have a barrel. Bat flies with wings. Wings are not barrels. Barrels are not wings. Bats can fly. There is a brown bat.

Query: The brown bat can fly

### 2) KB 2

Knowledge base is as follows:
Bat is a mammal.
Bat is used in baseball.
Bats have a barrel.
Bats have wings.
Bats can fly.
Flying requires wings.
Wings are not barrels.
Barrels are not wings.
Bat can be elegant.
There is a brown bat.

Query: Elegant Brown Bat has a barrel and can fly.

# Input prompts with knowledge base:

| Prompt 1 | Prompt 2 | Prompt 3 | Prompt 4 |
|---|---|---|---|
| This is the knowledge base:<br>Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bat flies with wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bats can fly.<br>There is a brown bat.<br>**The Brown Bat can fly.** using clausal forms prove by resolution refutation if this statement is true or not or inconclusive | This is the knowledge base:<br>Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bat flies with wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bats can fly.<br>There is a brown bat.<br>"Can we say Brown Bat can fly?"<br>Is this statement true or not or inconclusive? | Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bats have wings.<br>Bats can fly.<br>Flying requires wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bat can be elegant.<br>There is a brown bat.<br>**Elegant Brown Bat has a barrel and can fly.** using clausal normal forms prove by resolution refutation if this statement is true or not or inconclusive | Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bats have wings.<br>Bats can fly.<br>Flying requires wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bat can be elegant.<br>There is a brown bat.<br>**Elegant Brown Bat has a barrel and can fly.** Is this statement true or not or inconclusive? |
| Prompt 5 | Prompt 6 | Prompt 7 | Prompt 8 |

| | | | |
|---|---|---|---|
| Knowledge base is as follows:<br>Bat is a mammal. Bat is used in baseball. Bats have a barrel. Bat flies with wings. Wings are not barrels. Barrels are not wings. Bats can fly. There is a brown bat.<br><br>Query: The brown bat can fly<br><br>Steps: Generate First Order Logical form over knowledge base and query. Then convert to conjunctive normal forms. Then convert to skolemized clausal forms by using skolemization, and give entire knowledge in skolemized clausal normal forms including the query and negated query and then prove by resolution refutation if this statement is true or not or inconclusive | Knowledge base is as follows:<br>Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bats have wings.<br>Bats can fly.<br>Flying requires wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bat can be elegant.<br>There is a brown bat.<br><br>Query: Elegant Brown Bat has a barrel and can fly.<br><br>Steps: Generate First Order Logical form. Then convert to conjunctive normal forms. Then convert to skolemized clausal forms by using skolemization, and give entire knowledge in skolemized clausal normal forms including the query and negated query and then prove by resolution refutation if this statement is true or not or inconclusive | This is the knowledge base:<br>Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bat flies with wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bats can fly.<br>**There is a brown bat.**<br>"Can we say Brown Bat can fly?"<br>Is this statement true or not or inconclusive? | This is the knowledge base:<br>Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bats have wings.<br>Bats can fly.<br>Flying requires wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bat can be elegant.<br>**There is a brown bat.**<br>"Can we say Elegant Brown Bat has a barrel and can fly"? Is this statement true or not or inconclusive? |
| Prompt 9 | Prompt 10 | | |
| Knowledge base is as follows:<br><br>Bat is a mammal. Bat is used in baseball. Bats have a barrel. Bat flies with wings. Wings are not barrels. Barrels are not wings. Bats can fly. There is a brown bat.<br><br>Query: The brown bat can fly.<br><br>Generate one answer by following a two-stage reasoning.<br><br>Stage 1 : Generate First Order Logical form over knowledge base and query. Then convert to conjunctive normal forms. | Knowledge base is as follows:<br>Bat is a mammal.<br>Bat is used in baseball.<br>Bats have a barrel.<br>Bats have wings.<br>Bats can fly.<br>Flying requires wings.<br>Wings are not barrels.<br>Barrels are not wings.<br>Bat can be elegant.<br>There is a brown bat.<br><br>Query: Elegant brown bat has a barrel and can fly.<br><br>Generate one answer by following a two-stage reasoning.<br><br>Stage 1 : Generate First Order Logical form over knowledge base and query. Then convert to | | |

| | | | |
|---|---|---|---|
| Then convert to skolemized clausal forms by using skolemization, and give entire knowledge base in skolemized clausal normal forms including the query and negated query and then prove by resolution refutation if this statement is true or not or inconclusive.<br><br>In Stage 2, answer if there is any confusion between the meaning of context for two types of bats. Reason about "Can we say Brown Bat can fly?" Is this statement true or not or inconclusive? Without using Resolution refutation.<br><br>Answer:<br><br>1) Does stage 1 proof supply information about two types of bats in the knowledge base?<br><br>2) Does stage 2 proof supply information about the two types of bats in the knowledge base?<br><br>3) Does the conclusion in stage 1 contradict or support stage 2?<br><br>4) Is the final answer to the question "The brown bat can fly" inconclusive? | conjunctive normal forms. Then convert to skolemized clausal forms by using skolemization, and give entire knowledge base in skolemized clausal normal forms including the query and negated query and then prove by resolution refutation if this statement is true or not or inconclusive.<br><br>In Stage 2, answer if there is any confusion between the meaning of context for two types of bats. Reason about "Elegant brown bat has a barrel and can fly"? Is this statement true or not or inconclusive? Without using Resolution refutation.<br><br>Answer:<br>1) Does stage 1 proof supply information about two types of bats in the knowledge base?<br><br>2) Does stage 2 proof supply information about the two types of bats in the knowledge base?<br><br>3) Does the conclusion in stage 1 contradict or support stage 2?<br><br>4) Is the final answer to the question "Elegant brown bat has a barrel and can fly" inconclusive? | | |

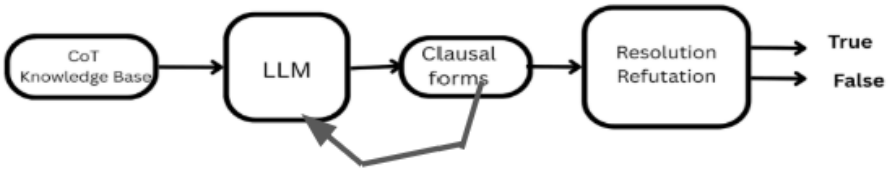
Prompt 5, Prompt 7 and Prompt 9 have the same KB with a different method to generate if a query is True or False or inconclusive. Prompts 6, 8 and 10 have the same KB each differing in the methods LLM needs to use to generate results if a query is True or False or inconclusive. Prompts 1 to 4 are designed to design more clear steps in the prompts, to assess what is working and what is not working and what is confusing the LLMs..

# Method summary:

Input will be processed and converted to Skolemized CNF, unify over CNF and apply resolution refutation. (Unify will be part of resolution refutation).

## Detailed Method Summary:

1) Input: knowledge base in natural language (as it was asked in the assignment 6 and 7)
2) Create a natural language knowledge base that can have multiple FOL representations, that is it can have multiple interpretations i.e. atleast 2.
3) Convert knowledge base to First Order Logic (FOL) representation 1 and 2 respectively (also without equality) (**using an existing work that I implemented a while ago**),
4) Generate Clausal normal forms from LLMs (claude 3.5, claude 3.7 and ChatGPT)
5) Apply Resolution refutation
6) Verify answers and compare with Classifical resolution refutation implementation




For each LLM, I gave a prompt in chat conversation User interface and response was generated by the LLM. For each one of this, to test the randomness of results, I repeated this process thrice for the same prompt within the same session. For this course project, I used max tokens set to 20000 and temperature set to 1, commonly for all LLMs.

# Output (same as output format from assignment 7):

Verify if LLMs are able to emit results of inconclusive where applicable by appealing to word sense disambiguation as well as world knowledge. Compare results between LLMs, Resolution Refutation implementation with that of correct answers.

For prompts that said with resolution refutation, we want LLMs to not preserve meanings as well as justify why it could be inconclusive due to word sense disambiguation.
For prompts without resolution refutation, LLMs conclude inconclusive wherever applicable.

## ClaudeAI and ChatGPT

ClaudeAI [6] and ChatGPT[5] are two large language models. ClaudeAI is built on ConstitutionalAI with self-reflection and feedback loop to learn from its responses to avoid generating harmful texts. ClaudeAI follows Reinforcement Learning with AI feedback (RLAIF) [6] while ChatGPT[4] follows Reinforcement Learning with Human Feedback (RLHF) [5]. Prompts are inputs to Large Language Models (LLMs) and they can be adjusted to get required output in specific formats. The LLMs are sensitive to prompts and tend to learn in-context [5]. Therefore applying prompt specific techniques to compare to that of traditional resolution refutation results would require consistent format and design of prompts.

## Intuition and Motivation behind this problem (based on review comments):

GPTx models have been vastly tested for First order logic forms and translation from natural language to first order logic. I could not find works that were tested on Claude and more recent versions of the GPTx, Claude and other LLMs, have been upgraded. They are known to generate nearly accurate First Order Logic forms, clausal forms. The explicit-reasoning tasks still require hand-crafted domain-specific rules and logic that are difficult to scale and generalize to other applications [23]. This also brings out the problem of lack of common consensus for measuring reasoning capabilities in LLMs [23].

## First Order Logic as a language to represent Natural Language:

Natural Languages are known to have inherent expressive power [1]. First Order Logic provides a mechanism for systems to better understand unstructured text towards language understanding tasks. [3] Capturing the expressive power of natural language by using First Order logic representations is another commonly followed direction [1] [2].

## Results:

For this project, I manually generated a knowledge base with word sense disambiguation over a single word which is available in the prompt table in the following page as well as with full derivations in the Example Knowledge Base

section. I tested this among claude-3.5-haiku20241022, claude-2.7-sonnet-20250219. Claude 3.5 and GPT.

- Resolution refutation sometimes sterminates but sometimes does not.
- I tested reasoning for true or false or inconclusive with resolution refutation and one without resolution refutation.
- I tested reasoning with both methods of implicit and explicit reasoning. I created the prompt to generate all steps required for Resolution Refutation using natural language knowledge base as well as one without. I called stage with RR as stage 1 and stage without RR as stage 2. I asked 4 questions, which are part of the answer, where the model makes use of both stages and reasons and answers the questions.

## Preliminary results for designing relevant, consistent and specific CoT prompts 1,2,3 and 4 :

This section only includes results for prompts 1 to 4 described in the Input Prompts section.

| Prompts | Claude 3.5 | ChatGPT | Claude 3.7 | Resolution Refutation | **Correct answers** |
|---|---|---|---|---|---|
| Prompt 1 With resolution refutation | True | True | Shows RR proof and also concludes Inconclusive with reasonable explanation | Terminates, True | True |
| Prompt 2 | True | Inconclusive | Inconclusive with reasonable explanation | Terminates, True | Inconclusive with word sense disambiguation |
| Prompt 3 With resolution refutation | Inconclusive | Inconclusive | False | Does not terminate | False, does not derive bottom |
| Prompt 4 Without resolution refutation | True | Inconclusive | True | Does not terminate | False with reasoning using word sense disambiguation |

## Final results for prompts 5,6,7 and 8:

Each prompt was fed to LLM thrice without any modification.
Results for each repeated prompt fed to LLM, are denoted as
True - T

False - F
Inconclusive - IC

Note: Prompts 5 and 6 are the only prompts that request Resolution Refutation.
Prompts 7 and 8 do not use Resolution Refutation, require common sense reasoning and word sense disambiguation from KB context.

Elegant Bat could be a mammal or bat in baseball. But bats cannot have a barrel and fly at the same time since wings != barrel. Both should be acceptable.

| KB | With RR YES/NO/ BOTH | Prompts | Claude 3.5 | Claude 3.7 | ChatGPT gpt-4o | ChatGPT Gpt-4o With "Think before responding" | ChatGPT 4o-mini (Advanced reasoning) | Resolution Refutation | **Correct answers** |
|---|---|---|---|---|---|---|---|---|---|
| 1 | YES | Prompt 5 | T, T, T | T, T, T | T, T, T | T, T, T | T, T, T | Terminates, resolves, satisfiable | **T** |
| 2 | YES | Prompt 6 | T, T, T | IC,T, F | T, T, T | T, T, T | T, T, T | Terminates, resolves, satisfiable | **F** |
| 1 | NO | Prompt 7 | T, T, T | IC, IC, IC | T, IC, IC | IC, IC, IC | T, IC, T | Same as prompt 5 | **IC** |
| 2 | NO | Prompt 8 | T, T, T | F, IC, F | T, IC, IC | IC, IC, IC | T, IC, IC | Same as prompt 6 | **IC** |
| 1 | BOTH | Prompt 9 | T, T, T | T, IC, IC | T, T, T | IC, IC, IC | IC, IC, IC | Same as prompt 5 | **IC** |
| 2 | BOTH | Prompt 10 | IC, IC, IC | F, F, F | IC, IC, IC | F, F, F | IC, IC, IC | Same as prompt 6 | **F** |

## Observations:

Based on the results from strictly using Resolution Refutation i.e. prompts 5 and 6, results vary over Claude 3.7. But for all other LLMs Claude 3.5, ChatGPT versions - GPT-4o, GPT-4o with Think before responding, 4o-mini, the results are consistent, specifically randomness is less. Due to the limited free-tier use for some LLMs, the test for randomness can be changed. The reason for variations in Claude 3.7, and small but initial variations in GPTx models in Chat was due to representations and signatures in First Order Logic forms that were generated initially. Depending on the number of predicates,

number of constants, the number of possible worlds i.e. number of interpretations would vary. Given that the skolemized clausal forms are not meaning-preserving which is a required feature for translating from Natural language to First Order Logic, traditional sense of NL to FOL is treated as symbolizing natural language sentences. Symbolizing would also introduce interpretations. Therefore, the same Natural language sentence can be interpreted in many ways i.e. the number of interpretations in First Order Logic varies. For example, whether to consider brown_bat as a constant or as a predicate are two variations observed between Claude-3.5-Haiku and Claude-3.7-sonnet respectively. Claude-3.7-sonnet recognized the word sense disambiguation and introduced Bat_animal and Bat_baseball as two different predicates, in one repetition of prompt-response interaction and in the other two repetitions of same prompt, Claude-3.7-sonnet introduced Bat, Mammal etc like other LLMs.
For Claude-3.5-haiku, most of the responses do not influence the conclusion even though word sense disambiguation is recognized.

In the prompts 9 and 10, where I combined prompt 5 with prompt 7 and prompt 6 with prompt 8, Claude-3.7-sonnet recognizes the ambiguity prior to generating the results for steps in RR method and non-RR methods respectively. The planning during the Thinking Phase before responding, generated strategy by addressing what is in the prompt. However in this case, to assist the LLMs with explicitly stating the obvious that there is inherent ambiguity due to word sense disambiguation. Ambiguity awareness, where the word meanings or the sentence contexts are not polar opposites to each other, can significantly introduce confusion and challenge the planning. The Thinking phase in LLMs does provide some context, but as the results show, it still does not suffice to state the obvious.

## Key Observations and questions/potential answers:

1) LLMs generated clausal forms for the KB2
   KB2

   Knowledge base is as follows:
   1. Bat is a mammal.
   2. Bat is used in baseball.
   3. Bats have a barrel.
   4. Bats have wings.
   5. Bats can fly.

6. Flying requires wings.
7. Wings are not barrels.
8. Barrels are not wings.
9. Bat can be elegant.
10. There is a brown bat.

11. Query: Elegant Brown Bat has a barrel and can fly.

Skolemized Clausal Normal forms generated by Claude 3.7 Sonnet that uses Thinking before responding:

1. ¬Bat_animal(x) ∨ Mammal(x)
2. ¬Bat_baseball(x) ∨ UsedInBaseball(x)
3. ¬Bat_baseball(x) ∨ HasBarrel(x)
4. ¬Bat_animal(x) ∨ HasWings(x)
5. ¬Bat_animal(x) ∨ CanFly(x)
6. ¬CanFly(x) ∨ HasWings(x)
7. ¬HasWings(x) ∨ ¬HasBarrel(x)
8. ¬HasBarrel(x) ∨ ¬HasWings(x)
9. Bat(c1)
10. Elegant(c1)
11. Bat(c2)
12. Brown(c2)
13. ¬Bat(x) ∨ Bat_animal(x) ∨ Bat_baseball(x)
14. ¬Bat_animal(x) ∨ Bat(x)
15. ¬Bat_baseball(x) ∨ Bat(x)

# Query and Negated Query

16. Query: Elegant(x), Brown(x), Bat(x), HasBarrel(x), CanFly(x))

**Observation**: LLM generates KB with word sense disambiguation (out of scope of KB) and world knowledge with common sense
**Following facts are not available in the KB:**
- Mammals are animals
- Bat is an animal

- Bat animal does not have a barrel
- Bat animal can fly
- Bat can imply Bat animal
- Bat can imply Bat baseball

- Do LLMs really use resolution refutation for the clausal forms they generated from Natural Language Knowledge bases?
- DO LLMs run Resolution refutation to derive the proof for resolution?
- Are LLMs able to "decide" a satisfiability of query using Resolution Refutation alone?
    - **Yes and no, yes they do on small KBs but it is unclear about larger KBs (document summarizations, though there are claims that they seem to do better)**
- Do LLMs "decide" a satisfiability of a query with word sense disambiguation and world knowledge outside of the given KB using Resolution Refutation alone?
    - **Whether we include in the prompt to use Resolution refutation or not, and whether we ask LLM to use word sense disambiguation or not, LLMs seem more likely to use word sense disambiguation as well as world knowledge and generated answers.**
- Do LLMs "decide" a satisfiability of a query with word sense disambiguation and world knowledge outside of the given KB WITHOUT using Resolution Refutation alone?
    - **This is still unclear, but may be solved like a human like and by trying to predict the next best token.**
- Do LLMs do better with best of both: First Order Logic combined with natural logic and common sense reasoning together?
    - **Prompts 9 and 10 demonstrate LLMs do better with combined approach, they are even able to generate some logical, world sense knowledge information.**

## Time and Space complexity, State Space

W.r.t to time, only one pair at a time is resolved, when there are multiple complementary literals and two statements can get illegal resolution depending on the sequence of resolution. The order matters. When we do hand-solve, we have some clues to make it more general, but it is difficult to stop the prover as the nested functions for a single predicate can grow. The worst case size can grow exponentially and the prover can run

forever. Size of history increases as the resolution explores all possible combinations. Space depends on the number of clauses and complexity of the clauses (depth of nested clause). We did not consider equality and it seems that it reduces the number of possibilities or paramodulation possibilities for this specific use case of considering equality.

Resolution is refutation complete, if a set of clauses, that are in first order clausal normal forms, are unsatisfiable, then the resolution derives empty clauses and verifies unsatisfiability. But Resolution is NOT refutation complete if the set of clauses may be satisfiable, that is there is a clause that is a logical consequence of the set of clauses, as it may never terminate. If C is a first order clausal normal form that is unsatisfiable, then $C \vdash \perp$, since resolution combined with skolemization and classification is a refutation complete calculus for First Order Logic.**[11]** By Lifting lemma and Herbrand's theorem, proof for First-order resolution's refutation completeness can be proved. If a formula is satisfiable, proof search can terminate or go one forever, as there are infinitely many clauses that can be generated by resolution. Therefore resolution is not a decision procedure for First order logic. There may be other forms of resolution [12] but we only use conventional skolemization decomposition for this project. The conventional skolemization decomposition (CSD) are not meaning-preserving clausal forms. [13] LLMs on the other hand, terminate and don't get stuck whether to conclude true, false or inconclusive, despite various other issues surrounding accuracy and planning during deductive reasoning. [7][15]

## How to generate a really long knowledge base and convert to Skolemized clausal forms?

1) Creating larger knowledge base and converting to clausal forms - Use LogicLLAMA and Claude AI to create a really large Knowledge base in clausal forms. And evaluating if the resulting clausal forms are correct may be challenging due to hallucinations and other known issues.[7]
2) Future work is focussed on generating dataset of knowledge bases with word sense disambiguation using LLMs and evaluate LLMs.

# Example Knowledge Bases, Clausal Forms and proofs

VS

## Example Knowledge base 1:

Bat is a mammal.
Bat is used in baseball.
Bats have a barrel.
Bat flies with wings.
Wings are not barrels.
Barrels are not wings.
Bats can fly.
There is a brown bat.
**--- query ---**
Brown Bat can fly.
**--- negated query ---**
Brown Bat cannot fly.

Will derive bottom since the negated query is False and the query is True.
There is at least one bat that is brown and bats can fly.

### Skolemized Clausal forms (manual):

1. Mammal(Bat) | Brown(Bat)
2. UsedIn(Bat, Baseball)
3. ( ∀x (Bat(x) -> Has(x, Barrel))
4. ( ∀x (Bat(x) -> ∃y (Wing(y) ∧ FliesWith(x,y) ) ) )
5. ( ∀x (Wing(x) -> ¬ Barrel(x) )
6. ( ∀x (Barrel(x) -> ¬ Wing(x) )
7. ( ∀x (Bat(x) -> CanFly(x))
8. ( ∃x (Bat(x) ∧ Brown(x) ) )
   --- query ---
   ( ∃x ( Bat(x) ∧ Brown(x) -> CanFly(x) ) )
   --- negated query ---
   ¬ ( ∃x ( Bat(x) ∧ Brown(x) -> CanFly(x) ) )

**query:**

( ∃x ( Bat(x) ∧ Brown(x) -> CanFly(x) ) ) => ∃x ( ¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x) ) )

∃x ( ¬ Bat(x) ∨ ¬ Brown(x)) ∨ CanFly(x) ) )

Skolemization constant c ¬ Bat(c) ∨ ¬ Brown(c) ∨ CanFly(c)

Final Clausal form **{** ¬ Bat(c) , ¬ Brown(c)), CanFly(c) ) ) **}**

**Negated query:**

( ∃x ( Bat(x) ∧ Brown(x) -> CanFly(x) ) ) => ¬ ∃x ( ¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x) ) )

Apply Quantifier Negation ∀x ¬ ( ¬ (Bat(x) ∧ Brown(x)) ∨ CanFly(x) ) )

∀x ( ¬¬ (Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x) ) )

∀x ( (Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x) ) )

( (Bat(x) ∧ Brown(x)) ∧ ¬CanFly(x) ) )

Final Clausal form **{** {Bat(x) }, { Brown(x) }, {¬CanFly(x) } **}**

**Proof:**

1. Mammal(Bat) | Brown(Bat)
2. UsedIn(Bat, Baseball)
3. ¬Bat(x) ∨ Has(x, Barrel)
4. ¬Bat(x) ∨ Wing(f(x))
5. ¬Bat(x) ∨ FliesWith(x,f(x))
6. ¬Wing(x) ∨ ¬Barrel(x)
7. ¬Barrel(x) ∨ ¬Wing(x)
8. ¬Bat(x) ∨ CanFly(x)
9. ¬Bat(x) ∨ Brown(x)
10. Bat(c)
11. Brown(c)
12. ¬CanFly(c)

Resolve 8 & 10 => ¬Bat(x) ∨ CanFly(x) and Substitute x = c

¬Bat(c) ∨ CanFly(c) Bat(c)

—----------

CanFly(c)

13. CanFly(c)

Resolve 12 & 13 => CanFly(c) ¬CanFly(c)

—----------

[ ]

**Therefore it entails that Brown Bat can fly.**
**Proof steps 4. Not meaning preserving.**

**Example Output for this problem is:**

1: ¬Bat(c) ∨ CanFly(c)
2: Bat(c)
3: CanFly(c)
4: ¬CanFly(c)
1 & 2 give 3: CanFly(c)
3 & 4 give 5: <empty>

## Knowledge base 2 (with noise):

Bat is a mammal.
Bat is used in baseball.
Bats have a barrel.
Bats have wings.
Bats can fly.
Flying requires wings.
Wings are not barrels.
Barrels are not wings.
Bat can be elegant.
There is a brown bat.
**--- query ---**
Elegant Brown Bat has a barrel and can fly.
**--- negated query ---**
Elegant Brown Bat does not have a barrel or cannot fly.

**Will not derive bottom since the negated query is true and the query is false.**
There is atleast one bat that is brown, bats can fly, flying requires wings and wings are not barrels and barrels are not wings.

## Skolemized Clausal forms (manual):

1. Mammal(Bat)
2. UsedIn(Bat, Baseball)
3. ¬Bat(x) ∨ Barrel(b(x))
4. ¬Bat(x) ∨ Has(x,b(x))
5. ¬Bat(x) ∨ Wing(w(x))
6. ¬Bat(x) ∨ Has(x,w(x))
7. ¬Bat(x) ∨ CanFly(x)
8. ¬CanFly(x) ∨ Wing(wf(x))
9. ¬CanFly(x) ∨ Has(x,wf(x))
10. ¬Wing(x) ∨ ¬Barrel(x)
11. ¬Barrel(x) ∨ ¬Wing(x)
12. Bat(e)
13. Elegant(e)
14. Bat(c)
15. Brown(c)
16. Bat(d)
17. Brown(d)
18. ¬CanFly(d)

Resolve 7 and 16
¬Bat(x) ∨ CanFly(x) and Bat(d)
Substitute x = d

¬Bat(d) ∨ CanFly(d)  Bat(d)
—----------
CanFly(d)

19. CanFly(d)

Resolve 19 and 18,

¬CanFly(d)  CanFly(d)
—----------
[ ]

**Example Output for this problem is:**

1: ¬Bat(x) ∨ CanFly(x)
2: Bat(d)
3: ¬CanFly(c)
4: CanFly(c)
1 & 2 give 3: ¬CanFly(c)
3 & 4 give 5: <empty>

**True.**

**CNF grammar, same as given in :**

clause → literal | clause
| literal
literal → predicate
| - predicate
predicate → capitalized-name ( term-list )
term-list → term , term-list
| term
term → capitalized-constant-name
| capitalized-function-name ( term-list )
| lowercase-variable-name

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29.

# Appendix

## Example problem to explain context for why conventional skolemized normal forms are NOT Meaning-preserving:

### Selecting example of Tax-cut problem from [4] :

1) Any person who has two children or more can get discounted tax
2) Peter has a child
3) Peter has a child named Paul
4) Query: Peter gets a tax cut
5) Negated Query: Peter doesn't get a tax cut

### First Order Logic form:

1) $F1$: $\forall x, y, z\ ((hasChild(x, y) \wedge hasChild(x, z) \wedge neq(y, z)) \rightarrow TaxCut(x))$
2) $F2$: $\exists x\ hasChild(Peter,\ x)$

3) $F3$: $hasChild(Peter,\ Paul)$
4) $Query$: $TaxCut(Peter)$
5) $Negated\ Query$: $\neg TaxCut(Peter)$

## CNF form:

1) $F1$: $\forall x, y, z\ ((hasChild(x, y) \wedge hasChild(x, z) \wedge neq(y, z)) \rightarrow TaxCut(x))$
2) $F2$: $\exists x\ hasChild(Peter,\ x)$
3) $F3$: $hasChild(Peter,\ Paul)$
4) $Query$: $TaxCut(Peter)$
5) $Negated\ Query$: $\neg TaxCut(Peter)$

## Conventional Skolemization:

1) $F1$: $\forall x, y, z\ ((hasChild(x, y) \wedge hasChild(x, z) \wedge neq(y, z)) \rightarrow TaxCut(x))$
   a) $(hasChild(x, y) \wedge hasChild(x, z) \wedge neq(y, z)) \rightarrow TaxCut(x)$
   b) $\neg\ (hasChild(x, y) \wedge hasChild(x, z) \wedge neq(y, z)) \vee TaxCut(x)$
   c) $(\neg hasChild(x, y) \vee \neg\ hasChild(x, z) \vee \neg\ neq(y, z)) \vee TaxCut(x))$
2) $F2$: $\exists x\ hasChild(Peter,\ x)$
   a) **Skolemize this:** $\exists x\ hasChild(Peter,\ x)$
      $hasChild(Peter,\ f1(x))$ where $f1$ is a skolem function
      **Result:** $hasChild(Peter,\ f1(x))$
3) $F3$: $hasChild(Peter,\ Paul)$
4) F4: $Negated\ Query$: $\neg TaxCut(Peter)$

## Resolution Refutation Solution using Conventional Skolemized Causal forms:

Skolemized clausal forms:

a) C1: $(\neg hasChild(x, y) \vee \neg\ hasChild(x, z) \vee \neg\ neq(y, z)) \vee TaxCut(x))$
b) $C2$: $hasChild(Peter,\ f1(x))$
c) $C3$: $hasChild(Peter,\ Paul)$
d) $C4$: $\neg TaxCut(Peter)$
e) $Query$: $TaxCut(Peter)$

Resolution Refutation Solution:

1) a) and b) =>
   1) Convert $(\neg hasChild(x, y) \vee \neg\ hasChild(x, z) \vee \neg\ neq(y, z)) \vee TaxCut(x))$ with x = Peter =>
      $C1$: $(\neg hasChild(Peter, y) \vee \neg\ hasChild(Peter, z) \vee \neg\ neq(y, z)) \vee TaxCut(Peter))$

2) Substitute y with $f1(a)$ in 1.1 C1 and replace x with 'a' in C2 a) =>
   $C1$: $(\neg hasChild(Peter, f1(a)) \vee \neg\, hasChild(Peter, z) \vee \neg\, neq(y, z)) \vee TaxCut(Peter))$
   $C2$: $hasChild(Peter,\ f1(a))$
   Resolve C1 and C2, resolvent is:
   $F5$: $(\neg\, hasChild(Peter, z) \vee \neg\, neq(y, z)) \vee TaxCut(Peter))$

2) c) and C5
   1) $C3$: $hasChild(Peter,\ Paul)$ and
      $C5$: $(\neg\, hasChild(Peter, z) \vee \neg\, neq(y, z)) \vee TaxCut(Peter))$
   2) Replace z with Paul in C5 =>
      $C5$: $(\neg\, hasChild(Peter, Paul) \vee \neg\, neq(y, Paul)) \vee TaxCut(Peter))$
   3) Resolve C5 and c), resolvent is: $C6$: $\neg\, neq(y, Paul)) \vee TaxCut(Peter)$

3) C4 and C6, resolvent is $C7$: $\neg\, neq(y, Paul))$ and not an empty set which proves original query $TaxCut(Peter)$ is True but is actually False since Peter does not meet the eligibility criteria for the Tax Cut.

$C7$: $\neg\, neq(y, Paul))$ bottom is not empty and therefore not derived. This is NOT an empty set. This is not the correct answer. We know the correct answer is $\neg TaxCut(Peter)$.

To address this problem, Existentially Quantified Conjunctive Normal Form ECNF is a proposed solution from [4].